Robotics, Artificial Intelligence
and Real-time Systems

Cognitive Systems: Neural Networks and Applied AI

Reinforcement Learning: Temporal Difference, SARSA(On-Policy), Q-Learning(Off-Policy)

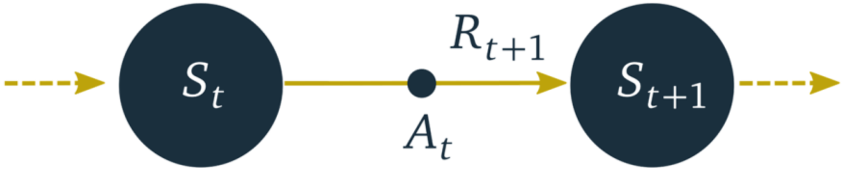
Prof. Dr.-Ing. habil. Alois Knoll
Mohammadhossein Malmir, M.Sc.

Agenda

- Temporal Difference (TD)
  - **TD Prediction: 1-Step TD**
- On-Policy and Off-Policy
- Temporal Difference (TD)
  - TD Control: SARSA (On-Policy)
  - TD Control: Q-Learning (Off-Policy)

TD Prediction

- **Temporal Difference** (TD) is a mixture of DP and MC that **samples** and **bootstraps**:



- In **TD prediction**, the Bellman equation is employed by iteratively updating value after every time step:

$$V(S_t) \leftarrow V(S_t) + \alpha \Big[\underbrace{\overbrace{R_{t+1} + \gamma V(S_{t+1})}^{\text{TD target}} - V(S_t)}_{\text{TD error}} \Big]$$

γ	discount rate	S	stochastical state function
A	stochastical action function	V	value estimates
R	stochastical reward function	α	learning rate

§Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, ***p.120.*** *Slide is adapted from CogSys21 RL lecture held by Prof. Dr.-Ing. Noah Klarmann.*

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On-Policy and Off-Policy

- Dilemma: Learning action values is conditional on subsequent *optimal* behavior, but behaving non-optimally is necessary in order to explore all actions and *find* the optimal ones.
- Question: How the agent can learn about the optimal policy while behaving according to exploratory policy? The answer is on off-policy learning.
- **Target policy** is the policy that the agent estimates its value function according to expected returns under that (what we have seen so far as $\pi(a|s)$).
- **Behavior policy** is the policy that the agent actually behave according to it to sample actions within the interaction with the environment (denoted by $b(a|s)$).

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On-Policy and Off-Policy

- The agent is learning **on-policy** when the behavior policy and target policy are **the same**.
- The agent is learning **off-policy** when the behavior policy **differs from** the target policy.
- On-policy learning is actually a compromise as it learns action values not for the optimal policy, but for a near-optimal policy that still explores.
- By separating the optimal target policy from the exploratory behavior policy, off-policy learning helps in dealing with the exploration problem.
- In order to have valid off-policy learning, the chosen behavior policy must cover the target policy:

$$\pi(a|s) > 0 \quad \text{where} \quad b(a|s) > 0$$

a	particular action	π	target policy
s	particular state	b	behavior policy

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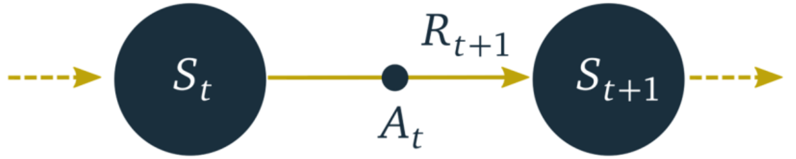
TD Control

- In **TD control**, state-action pairs are updated while acting greedily on the value function:



$$\Delta Q(S_t, A_t) = ? \qquad Q(S_{t+1}, A_{t+1})$$

$$\underbrace{R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})}_{\text{target}}$$

- on-policy TD control (SARSA):

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left[\underbrace{R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})}_{\text{on-policy target}} - Q(S_t, A_t)\right]$$

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γ	discount rate	S	stochastical state function	q	value for a specific action
A	stochastical action function	R	stochastical reward function	Q	value estimates for a specific action

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On-Policy vs. Off-Policy TD Control

- SARSA (**on-policy**, TD control):

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\big[\underbrace{R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})}_{\text{on-policy target}} - Q(S_t, A_t)\big]$$

- Q-Learning (**off-policy**, TD control)

following a further non-exploring greedy policy

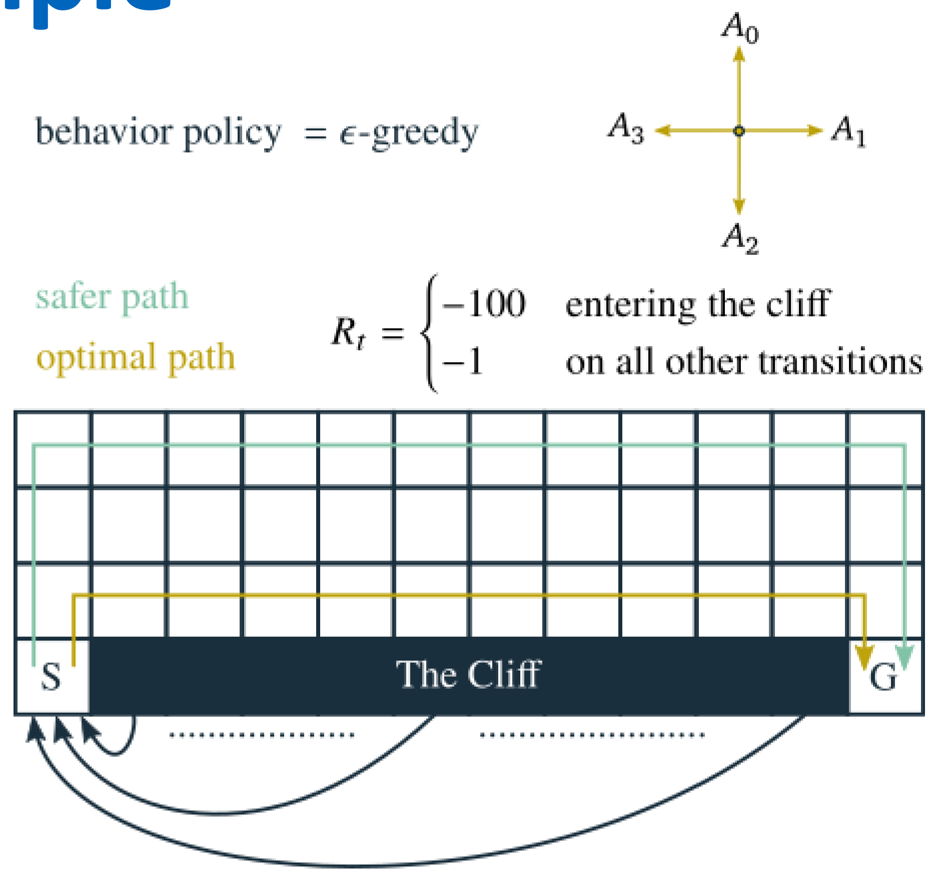
$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\big[\underbrace{R_{t+1} + \gamma \max_a Q(S_{t+1}, \overset{\downarrow}{a})}_{\text{off-policy target}} - Q(S_t, A_t)\big]$$

α	learning rate	Q	value estimates for a specific action
γ	discount rate	R	stochastical reward function
A	stochastical action function	S	stochastical state function

[§]*Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018,* ***p.129,131.*** *Slide is adapted from CogSys21 RL lecture held by Prof. Dr.-Ing. Noah Klarmann.*

TD Control: Cliff Walking Example

- What to consider when dealing with an off-policy method?
- Since Q-learning learns values for the optimal policy, it results in choosing actions that go through the optimal path.
- However, this translates into lower total returns due to having higher chance of falling to the cliff caused by the ε-greedy action selection.
- In contrast, since SARSA takes the action selection into account, it learns the longer but safer path.
- Hence, by learning on-policy SARSA achieves higher online performance than that of Q-learning.



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